UNIVERSITY OF MANITOBA
Department of Mathematics

COURSE: *Math 1210* EXAMINER: *Romain Branchereau*
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EXAM: *Final exam*

Grading table

Question:	1	2	3	4	5	6	7	8	9	10	Total
Points:	8	6	10	14	9	7	6	12	7	15	94
Score:											

1. (8 points) Prove by induction that

$$\begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^n = \begin{pmatrix} 1-2n & n \\ -4n & 1+2n \end{pmatrix}$$

for any $n \geq 1$.

Solution: Let us denote by $P(n)$ the statement

$$\begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^n = \begin{pmatrix} 1-2n & n \\ -4n & 1+2n \end{pmatrix}.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is $\begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^1 = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}$ and equal to the right hand side $\begin{pmatrix} 1-2\cdot 1 & 1 \\ -4\cdot 1 & 1+2\cdot 1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}$. Hence $P(n)$ is true for $n = 1$. **Induction step:** We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$\begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^k = \begin{pmatrix} 1-2k & k \\ -4k & 1+2k \end{pmatrix}. \tag{1}$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$\begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^{k+1} = \begin{pmatrix} -1-2k & k+1 \\ -4k-4 & 3+2k \end{pmatrix}. \tag{2}$$

Using the induction hypothesis (1), the left-hand side of (2) is

$$\begin{aligned}
\begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^{k+1} &= \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^{k} \\
&= \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} 1-2k & k \\ -4k & 1+2k \end{pmatrix} \\
&= \begin{pmatrix} -4k-(1-2k) & (2k+1)-k \\ 3(-4k)-4(1-2k) & 3(2k+1)-4k \end{pmatrix} \\
&= \begin{pmatrix} -1-2k & k+1 \\ -4k-4 & 3+2k. \end{pmatrix}
\end{aligned}$$

Hence, the proposition $P(k+1)$ is valid under the assumption that $P(k)$ is valid.
Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

2. (6 points) Find the Cartesian form of

$$z = \frac{i^{62}(\sqrt{2}+\sqrt{6}i)^5}{2^6(-\sqrt{6}-\sqrt{2}i)}.$$

Simplify as much as possible.

Solution: Using that $i^{62} = (i^2)^{31} = (-1)^{-31} = -1$ we find that

$$\begin{aligned}
z &= \frac{i^{62}(\sqrt{2}+\sqrt{6}i)^5}{2^6(-\sqrt{6}-\sqrt{2}i)} \\
&= \frac{(\sqrt{2}+\sqrt{6}i)^5}{2^6(\sqrt{6}+\sqrt{2}i)}. \qquad (3)
\end{aligned}$$

The exponential form of $\sqrt{2}+\sqrt{6}i$ is

$$\sqrt{2}+\sqrt{6}i = \sqrt{2}^3 e^{i\pi/3}.$$

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Hence

$$\begin{aligned}
z &= \frac{(\sqrt{2}^3 e^{i\pi/3})^5}{2^6 \sqrt{2}^3 e^{i\pi/6}} \\
&= \frac{1}{2^6} \sqrt{2}^{12} e^{5i\pi/3} e^{-i\pi/6} \\
&= e^{9i\pi/6} \\
&= e^{3i\pi/2} \\
&= -i.
\end{aligned}$$

3. Consider the polynomial $P(x) = x^3 + 4x^2 + 7x + 6$.

(a) (4 points) Use the rational root theorem to list all possible rational roots. Simplify the list as much possible using the other theorems that we have seen in class.

Solution: By the rational root theorem, if p/q is a rational root, then p divides 6 and q divides 1. So the possible rational roots are

$$p/q = \pm 1, \pm 2, \pm 3, \pm 6.$$

By Descartes rule of sign, since $P(x)$ has no sign changes there are no positive real roots. Hence we can reduce the list of possible rational roots to

$$p/q = -1, -2, -3, -6.$$

(b) (6 points) Find all the zeros of $P(x)$.

Solution: We can use the result of part (a) to do trial and error. The value -1 is not a root, but -2 is a root since

$$P(-2) = (-2)^3 + 4(-2)^2 + 7(-2) + 6 = -8 + 16 - 14 + 6 = 0.$$

Hence we can factor out $(x + 2)$:

$$\begin{array}{r|l} & \quad x^2 + 2x + 3 \\ \hline x+2 & \quad x^3 + 4x^2 + 7x + 6 \\ & -x^3 - 2x^2 \\ \cline{2-2} & \qquad 2x^2 + 7x \\ & \qquad -2x^2 - 4x \\ \cline{2-2} & \qquad\qquad 3x + 6 \\ & \qquad\qquad -3x - 6 \\ \cline{2-2} & \qquad\qquad\qquad 0 \end{array}$$

and conclude that

$$P(x) = (x^2 + 2x + 3)(x + 2).$$

The two roots of $x^2 + 2x + 3 = 0$ are $-2 \pm i\sqrt{2}$. It follows that the zeros of $P(x)$ are

$$-2 - i\sqrt{2}, \quad -2 + i\sqrt{2} \quad \text{and} \quad -2.$$

4. (*Part (d) is on the next page*) - Consider the vectors $\mathbf{u} = \langle 1, m, 2\rangle, \mathbf{v} = \langle m, 2m, 2\rangle$ and $\mathbf{w} = \langle 0, -3, 1\rangle$ where m is a real number.

(a) (2 points) For which value(s) of m are $\mathbf{u}$ and $\mathbf{u} \times \mathbf{v}$ perpendicular?

Solution: The cross product of two vectors is always perpendicular to the two vectors. Hence $\mathbf{u}$ and $\mathbf{u} \times \mathbf{v}$ are perpendicular for any value of m.

(b) (2 points) For which value(s) of m are $\mathbf{v}$ and $\mathbf{w}$ perpendicular?

Solution: They are perpendicular iff the dot product

$$\mathbf{v} \cdot \mathbf{w} = -6m + 2$$

is zero. That is when $m = 1/3$.

(c) (4 points) For which value(s) of m are the three vectors linearly dependent?

Solution: They are linearly dependent if and only if the determinant

$$\begin{vmatrix} 1 & m & 0 \\ m & 2m & -3 \\ 2 & 2 & 1 \end{vmatrix} = -(-3)\begin{vmatrix} 1 & m \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 1 & m \\ m & 2m \end{vmatrix} = -m^2 - 4m + 6$$

is zero. The roots of $m^2 + 4m - 6 = 0$ are $-2 \pm \sqrt{10}$, hence the vectors are linearly dependent for $m = -2 - \sqrt{10}$ and $m = -2 + \sqrt{10}$.

(d) (6 points) Now take $m = 1$. Write the vector $\langle 1, -12, 5\rangle$ as a linear combination of $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$.

Solution: When $m = 1$ the three vectors are $\mathbf{u} = \langle 1, 1, 2\rangle$,$\mathbf{v} = \langle 1, 2, 2\rangle$ and $\mathbf{w} = \langle 0, -3, 1\rangle$. Writing $\langle 1, -12, 5\rangle$ as a linear combination of $\mathbf{u}$,$\mathbf{v}$,$\mathbf{w}$ requires to find C_1, C_2, C_3 such that

$$C_1\langle 1,1,2\rangle + C_2\langle 1,2,2\rangle + C_3\langle 0,-3,1\rangle = \langle 1,-12,5\rangle$$
$$\iff \langle C_1 + C_2, C_1 + 2C_2 + 3C_3, 2C_1 + 2C_2 - C_3\rangle = \langle 1, -12, 5\rangle.$$

Since the equations must hold componentwise, we can write the equation as a nonhomogeneous system

$$\begin{aligned} C_1 + C_2 &= 1 \\ C_1 + 2C_2 - 3C_3 &= -12 \\ 2C_1 + 2C_2 + C_3 &= 5. \end{aligned}$$

Let us now solve the system to determine the solution (C_1, C_2, C_3). We bring the augmented matrix in RREF by Gauss-Jordan elimination:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 1 & 2 & -3 & -12 \\ 2 & 2 & 1 & 5 \end{array}\right) \begin{array}{l} \\ R_2 \longrightarrow R_2 - R_1 \\ R_3 \longrightarrow R_3 - 2R_1 \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & -3 & -13 \\ 0 & 0 & 1 & 3 \end{array}\right) R_2 \longrightarrow R_2 + 3R_3$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right) \begin{array}{l} R_1 \longrightarrow R_1 - R_2 \\ \\ \\ \end{array}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right)_{RREF} .$$

It follows that the unique solution is $(C_1, C_2, C_3) = (5, -4, 3)$ and

$$\langle 1, -12, 5\rangle = 5\mathbf{u} - 4\mathbf{v} + 3\mathbf{w}.$$

5. Consider the plane $\Pi\colon 2x - 3y + z = 5$, the line $L\colon x = 1 + t,\ y = 2 + t,\ z = 9 + t$ and the point $P(1, -1, 2)$.

(a) (3 points) Show that the line L is contained in the plane Π.

Solution: To verify if the line is contained in the plane, me must verify that the parametric equation satisfy the plane equation. Since

$$2(1 + t) - 3(2 + t) + (9 - t) = 2 + 2t - 6 - 3t + 9 - t = 5,$$

the line L is contained in the plane Π.

Alternatively, you can also verify that the vector $\langle 1,1,1\rangle$ parallel to the line L is perpendicular to the normal vector $\langle 2,-3,1\rangle$ of the plane Π. Since the dot product

$$\langle 1,1,1\rangle \cdot \langle 2,-3,1\rangle = 2-3+1=0$$

is zero, the vectors are perpendicular. Hence, the line L is contained in the plane Π.

(b) (6 points) Find an equation of the plane passing through the point P and intersecting the plane Π along the line L.

Solution: The vector $\mathbf{v} = \langle 1,1,1\rangle$ is parallel to the line. Moreover, setting $t=0$ we find that $Q(1,2,9)$ is a point on the line. Hence, the plane that we are looking for is parallel to the two vectors $\mathbf{v}$ and $\mathbf{PQ} = \langle 0,3,7\rangle$. Hence, a normal vector of the plane is

$$\mathbf{v}\times\mathbf{PQ} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 0 & 3 & 7 \end{vmatrix} = \mathbf{i}\begin{vmatrix} 1 & 1 \\ 3 & 7 \end{vmatrix} - \mathbf{j}\begin{vmatrix} 1 & 1 \\ 0 & 7 \end{vmatrix} + \mathbf{k}\begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} = 4\mathbf{i} - 7\mathbf{j} + 3\mathbf{k} = \langle 4,-7,3\rangle.$$

So an equation of the plane through P and the given line is

$$4(x-1)-7(y+1)+3(z-2)=0,$$

or equivalently

$$4x-7y+3z=17.$$

6. (7 points) Consider the system

$$\begin{aligned} x-y+2z &= 1 \\ x+2y+z &= -1 \\ 2x-y+3z &= 0. \end{aligned}$$

It is given to you that the system has a unique solution (x,y,z). Use Cramer's rule to find z only. (You do **not** need to find x and y).

Solution: The coefficient matrix of the system is

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 2 & 1 \\ 2 & -1 & 3 \end{pmatrix}.$$

By Cramer's rule, the entry z of the solution is given by

$$z = \frac{|A_3|}{|A|}$$

where

$$A_3 = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & -1 & 0 \end{pmatrix}.$$

By expanding along the first row we find that the determinant of A is

$$|A| = \begin{vmatrix} 1 & -1 & 2 \\ 1 & 2 & 1 \\ 2 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} - \begin{vmatrix} -1 & 2 \\ -1 & 3 \end{vmatrix} + 2\begin{vmatrix} -1 & 2 \\ 2 & 1 \end{vmatrix} = -2.$$

By expanding along the first row again we also find that the determinant of A_3 is

$$|A_3| = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & -1 & 0 \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ -1 & 0 \end{vmatrix} - \begin{vmatrix} -1 & 1 \\ -1 & 0 \end{vmatrix} + 2 \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} = -4.$$

Hence

$$z = \frac{|A_3|}{|A|} = \frac{-4}{-2} = 2.$$

7. (6 points) Consider the invertible matrix

$$A = \begin{pmatrix} 2 & -1 & -3 \\ 2 & 0 & 1 \\ 1 & -2 & -1 \end{pmatrix}.$$

Use the adjoint method to find the entry in the second column and third row of A^{-1}.

Solution: By the adjoint method

$$A^{-1} = \frac{1}{|A|} \operatorname{adj}(A) = \frac{1}{|A|} C^T = \frac{1}{|A|} \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix}$$

where

$$C = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

is the matrix of cofactors of A. The entry in the second column and third row of A^{-1} is

$$\frac{c_{23}}{|A|}.$$

The cofactor c_{23} of A is given by

$$c_{23} = -m_{23} = -\begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} = 3.$$

By expanding along the second row we find that the determinant of A is

$$\begin{vmatrix} 2 & -1 & -3 \\ 2 & 0 & 1 \\ 1 & -2 & -1 \end{vmatrix} = -2 \begin{vmatrix} -1 & -3 \\ -2 & -1 \end{vmatrix} - \begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} = 13.$$

The entry in the second column and third row of A^{-1} is $\frac{3}{13}$.

8. Consider the system

$$\begin{aligned}-x_1 + x_2 - 3x_3 + x_4 &= a\\ -x_1 + 2x_2 - 4x_3 + 2x_4 &= a^2\\ x_1 + 2x_3 &= a^3\end{aligned}$$

where a is a real number.

(a) (3 points) For which value(s) of a does the system have a unique solution?

Solution: For any value a, the system has $n = 4$ unknown and $m = 3$ equations. Since the rank r must satisfy $r \leq \min(3, 4) = 3$, we can never have $r = n = 4$, that is required to have unique solution. Hence there are no values of a for which the system has a unique solution.

(b) (9 points) Use Gauss-Jordan elimination to find the basic solutions of the system when $a = 0$.

Solution: We bring the augmented matrix in RREF:

$$\left(\begin{array}{cccc|c} -1 & 1 & -3 & 1 & 0\\ -1 & 2 & -4 & 2 & 0\\ 1 & 0 & 2 & 0 & 0\end{array}\right)\begin{array}{l} R_1 \to R_3\\ \\ R_3 \to R_1\end{array} \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0\\ -1 & 1 & -3 & 1 & 0\\ -1 & 2 & -4 & 2 & 0\end{array}\right)\begin{array}{l} \\ R_2 \to R_2 + R_1\\ R_3 \to R_3 + R_1\end{array}$$

$$\longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0\\ 0 & 1 & -1 & 1 & 0\\ 0 & 2 & -2 & 2 & 0\end{array}\right)\begin{array}{l} \\ \\ R_3 \to R_3 - 2R_1\end{array}$$

$$\longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0\\ 0 & 1 & -1 & 1 & 0\\ 0 & 0 & 0 & 0 & 0\end{array}\right).$$

Hence $x_1 + 2x_3 = 0 = 0$ and $x_2 - x_3 + x_4 = 0$, equivalently $x_1 = -2x_3$ and $x_2 = x_3 - x_4$. Setting the parameters $u = x_3$ and $t = x_4$ we find that the general solution is

$$\begin{pmatrix} x_1\\ x_2\\ x_3\\ x_4\end{pmatrix} = \begin{pmatrix} -2u\\ u - t\\ u\\ t\end{pmatrix} = u\begin{pmatrix} -2\\ 1\\ 1\\ 0\end{pmatrix} + t\begin{pmatrix} 0\\ -1\\ 0\\ 1\end{pmatrix}.$$

The two basic solutions are

$$\begin{pmatrix} -2\\ 1\\ 1\\ 0\end{pmatrix}, \quad \begin{pmatrix} 0\\ -1\\ 0\\ 1\end{pmatrix}.$$

9. (7 points) Consider the linear transformation $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ sending $\langle 3, 2\rangle$ to $\langle 8, 0\rangle$, and $\langle 1, 0\rangle$ to $\langle 2, -2\rangle$. First, use that T is linear to find the image of $\langle 0, 1\rangle$. Then, find the associated matrix.

Solution: We know that $T\langle 3, 2\rangle = \langle 8, 0\rangle$ and $T\langle 1, 0\rangle = \langle 2, -2\rangle$. Moreover, we have

$$\langle 3, 2\rangle - 3\langle 1, 0\rangle = \langle 0, 2\rangle = 2\langle 0, 1\rangle.$$

hence

$$2T\langle 0,1\rangle = T\langle 3,2\rangle - 3T\langle 1,0\rangle = \langle 8,0\rangle - 3\langle 2,-2\rangle = \langle 2,6\rangle.$$

After dividing both sides by T we find that $T\langle 0,1\rangle = \langle 1,3\rangle$. The columns of the associated matrix are exactly $T\langle 1,0\rangle$ and $T\langle 0,1\rangle$. Hence, the associated matrix is

$$\begin{pmatrix} 2 & 1 \\ -2 & 3 \end{pmatrix}.$$

10. (*Part (c) is on the next page*) - Consider the linear transformation T with associated matrix

$$A = \begin{pmatrix} 3 & 1 & 5 \\ 1 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}.$$

(a) (2 points) Find the image of $\langle 2,-3,1\rangle$ by T.

Solution: We have

$$T\langle 2,-3,1\rangle = \begin{pmatrix} 3 & 1 & 5 \\ 1 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ -1 \end{pmatrix}.$$

Hence, $T\langle 2,-3,1\rangle = \langle 8,-2,-1\rangle$.

(b) (7 points) Find the eigenvectors to the eigenvalue $\lambda = 2$.

Solution: The eigenvectors $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ to the eigenvalue $\lambda = 2$ are the nonzero solution of the homogeneous system $(A - 2I_3)\mathbf{v} = 0$. We have

$$A - 2I_3 = \begin{pmatrix} 1 & 1 & 5 \\ 1 & 1 & 5 \\ 0 & 0 & -3 \end{pmatrix}.$$

We bring the augmented matrix in RREF using Gauss-Jordan elimination:

$$\left(\begin{array}{ccc|c} 1 & 1 & 5 & 0 \\ 1 & 1 & 5 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right) \begin{array}{l} \\ R_2 \to R_2 - R_1 \\ R_3 \to -R_3/3 \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \begin{array}{l} R_1 \to R_1 - 5R_3 \\ \\ \\ \end{array}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \begin{array}{l} \\ \\ R_3 \leftrightarrow R_2 \end{array}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \begin{array}{l} \\ \\ RREF \end{array}.$$

Hence $v_3 = 0$ and $v_1 + v_2 = 0$, equivalently $v_1 = -v_2$. Setting $v_2 = t$, we find that the eigenvectors to the eigenvalue $\lambda = 2$ are

$$\mathbf{v} = \langle -t, t, 0\rangle = t\langle -1, 1, 0\rangle$$

where t is a nonzero real number.

(c) (6 points) Find the other eigenvalues. You do **not** need to find the corresponding eigenvectors.

Solution: The eigenvalues are the roots of the characteristic equation $|A-\lambda I_3| = 0$. We compute

$$\begin{aligned} |A-\lambda I_3| &= \begin{vmatrix} 3-\lambda & 1 & 5 \\ 1 & 3-\lambda & 5 \\ 0 & 0 & -1-\lambda \end{vmatrix} \\ &= -(1+\lambda)\begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} \\ &= -(1+\lambda)(2-\lambda)(4-\lambda). \end{aligned}$$

Hence the three eigenvalues are $\lambda = -1$, $\lambda = 2$ and $\lambda = 4$.